

ActInf Textbook Group ~ Cohort 2 ~ Meeting 6

(Chapter 3, part 1)

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okay hello everyone it's October 7th 2022 we're in the sixth meeting for the second cohort of the textbook group today we're going to be discussing chapter three for the first time so we have a few questions to explore on three but before we do would anyone like to just say hello or give any overview thoughts on chapter three before we go into more specific questions you can raise your hand or just unmute and just go for it I'll wait for someone to make a comment on chapter three yep to active inference with the principle of least action and the introduction of Markov blankets okay how does that differ from previous approaches or relative to what but I guess instead of starting with um instead of starting with the notion that you have some agent that's creating a representation of its world you start with like in axiomatic view from physics that when applied to biological entities GIF leads to the at least the same conclusion as chapter two all right awesome thank you anyone can raise their hand or just go for it General thoughts on chapter three and or what is the high road to active inference foreign having read chapters 2 and 3. how does anybody want to remark on figure 1.2 okay I'll make a short remark to me this figure I guess these uh elements like surprise minimization Mark of blanket looks like they are they are inserted in the pictures like sequentially but it doesn't seem to me that they actually represent a sequence for understanding the the the approach they're just various aspects of uh of either the the physics approach or in the lower part of the figure of the Bayesian approach if I if I understood it correctly because at first I thought it was like a path you know starting from the free energy principle and then first to self-organization then surprise minimization Mark of blanket predictive processing and I don't know if it's if it is actually a progression of Concepts or actually I don't know what do you think yeah it's a great question anyone can give a thought what do the ordering of the terms mean are these train stops along the way blue or child so I just wanted to say that um the live stream where Maria really gave a lot of background on predictive processing predictive coding like the difference between these two might be super

helpful um with regard to like a progression or just evaluating that lineage in more detail um yeah sure I'll add link there um one option would be that these are historical like this is this chronology of ideas another one would be that they're pedagogical that this is a learning path another might be uh formal or technical like one of them are one of the directions are dependencies or required but not necessarily the most pedagogical path Ali [Music] yeah I also think that the organization of this uh particular figure is more like the branch because of a tree and then uh something uh sequential so they might not represent any uh specific preference to go through all of these steps in order to achieve a particular goal where just try it just tries to lump together relevant Concepts in two different branches so that's why I see it at least yes the tree um fep tree from uh 2010 with a lot of the early sort of directions and pointers towards areas of synthesis anyone else want to just add any other thoughts like what is the high road or how did they feel about chapter three overall Technical and specific questions to dive into but it's really important to understand chapter two and the low road and chapter 3 being the high road and the way that they're distinguished in terms of the low road beginning from a framework for doing statistics learning updating and so on but not necessarily an imperative for survival in contrast in the high road the beginning stance is around the imperative for survival which is kind of a biological way to frame it or the slightly even more deflationary imperative for repeated measurements which is more of a Quantum or a physics way to be framing the issue in terms of stochastic processes and measurement Theory where if something cannot be measuring itself it's not going to see itself around for long and if something is not a thing to something else it's not being measured over a given time scale within a given variance Criterion it also is going to be quote not a thing so this imperative for uh persistence and finding manifolds of stability or bounded spaces is an imperative that isn't intrinsically within Bayesian theorem but active inference is being situated at the intersection of the Bayesian methods and the imperative for persistence and or recur or repeated measurements Ali uh yes one of the I think the most probably the most Central concept of chapter three uh on which all the other material seems to um seems to evolve is the concept of Mark of blanket but but in this particular chapter I think the marker blanket I mean the nature of markup blanket has not made uh quite explicit so I just wanted to emphasize a critical point about this whole concept which has caused a lot of lots of confusions and misunderstandings around it and namely uh some people think markup blanket is actually some a kind of spatial temporal boundary around the system but as the paper and Bayesian mechanics makes clear uh marker blanket is actually a boundary a boundary on state space and it doesn't necessarily or it

doesn't always translate into any spatial temporal boundary in some cases uh first for example in the cells or organelles or things like that we can see the spatial temporal manifestation of markup blanket but that's not always the case so yeah that's one of the points I think is crucial to understanding this whole High Road approach to active inference hmm thank you Ali very interesting points so they do in the first box of the chapter and in the first non-introduction section discuss Marco blankets so it's definitely something that we want to um follow on so let's we're done for now with the just general discussion around what the high road is um good yes Giuseppe please okay no it's just on this topic that it's true that the mark of blanket is explained they quite clearly if it's succinctly in that part but then I don't think it throughout the book is actually uh is actually used or describe the in details from the computational point of view that I mean I think it's it's described it's it's defined there but then I to me since that doesn't appear much in the in the more computational later parts of the book that that was my impression so the concept is quite clear uh but I'm not sure how it um it applies then practically just just laying out the model probably in laying out the specifics of the model that defines by itself the the relevant Mark of blanket probably thank you um Ali uh yes uh one reason for that might be at least according to my personal communication with Maxwell ramstead uh you see uh the problem of identifying markup blankets from data uh is still an outstanding uh issue and problem uh among fep researchers uh on which versus lab is actively working on so that's not a resolved problem I guess uh yet so that's I think one of the reasons they have not alluded to the concept of Market blankets so much throughout the whole through the throughout the rest of the book uh the other reason is uh the recent literature somehow tries to uh reduce its dependence on this whole concept of markup blanket um because of various reasons so yeah these two reasons might be sufficient to explain its lack of appearance or at least explicit appearance in much of the book Thank you great yeah I think um this this question will help us dive further into this question on page 43 they uh are writing so first off anyone who wanted to describe on a more technical note we can bring it up later or sooner what a Markov blanket is because it is one of the core terms in brief a markup blanket is the set of variables that mediate all statistical interactions between a system and environments and those parentheses cover a lot that is not the same thing as the actual mechanical or material interactions between the system and the environments that is the map territory fallacy or failure to distinguish although one can't also swing too far into the map territory fallacy fallacy area which we've talked about earlier this the markout blanket and what it does is a restatement of the classical action perception cycle where an Adaptive system

and its environment can interact only through actions and observations respectively so how's the Markov blanket a restatement of the classical action perception cycle I feel that this is an important question to understand to contextualize active inference Within the history of models like cybernetics control theory ecological psychology and qualitative and quantitative models of perception and action kind of goes without saying but it's a really important note and we see multiple representations of the action perception Loop so figure three one shows the action perception Loop along with the particular partition that's the one that's separated into internal and external States along with the blanket States which makes internal and external conditionally independent and then the blanket states are further decomposed into active States those with outgoing dependencies from the perspective of internal States and sensory States those with internal dependencies although note some asterisks in the interpretation for example around which arrows have two directions and which ones don't so here's figure 3.1 the Markov blanket it's Dynamic it's changing in time the states are changing in time separating partitioning internal States here the brain from the environments and all of the functions are described in terms of rates of change $\dot{x}$ $\dot{u}$ $\dot{y}$ and $\dot{\mu}$ and the dot over the top is like the derivative so they're all being solved for in terms of how they change their time and they're all being described with these F functions that are themselves comprised of a deterministic component that's the $\bar{u}$ average rate of change over these arguments coming into the deterministic components of the equation and Ω which is some sort of random fluctuation parameter so it's kind of like a signal to noise partitioning statistical variance Within the change characteristics of a given state corresponding to particular partitioned variables but we've also seen other representations of the action perception Loop so here's one of the simpler ones from figure one this is the first figure in the book and we can think about how like the active States even though they're coming up from the right here and they're coming kind of down from the left here in both situations it's like this is the engagement between the agent and the niche okay but then there are some other representations too like figure two two here we see a proto-blanket forming though not described as such and without like for example an arrow connecting these two nodes between observation and action and more of an emphasis on the way that hidden states are being either actually generative of the observations that's the generative process or that hidden states of the generative model are doing inference on the observations so that's one other $\bar{u}$ representation and then there's this representation where here action is coming out of discrepancy as is perception learning generative model change and we see observation coming into what is the action perception

cycle and where do Markov blankets come into play in terms of even the classical ways of thinking about control systems if one looks up control Loop they'll find all kinds of diagrams from cybernetics and control theory that can entirely be interpreted within a sort of entity getting input doing some sort of processing and having outputs how is the Marco blanket helping us approach this Nexus of the entity in the niche well one thought or important piece to note is that some blanket States will exist for any Bayesian graph that isn't fully connected anytime that you have a non-fully connected base graph so not every node is connected to every node then there will be some other nodes upon which two other sets are a conditionally independent that's just what the nodes and edges mean in this kind of a graphical model which is if nodes are random variables and edges are statistical relationships then nodes directly connected by an edge have a direct statistical relationship again not implying anything about the material or the causal nature of the so then if two nodes do not have a statistical relationship they're either part of two different graphs which can be seen as basically partitions from each other or they're going to be part of a connected component of a graph but there's some other nodes one or more that intermediate such that those two nodes of Interest which are arbitrarily called internal and external have conditional independencies that ends up simplifying and being important for a lot of the computation and it respects and reflects the sparse coupling of features in the world so it's always just important to like think back like where is something new coming in to active inference people have been using partially observable Markov decision processes for many years for perception and control issues before active inference those models also are imbued with Markov blankets just like any other markovian setting in Time series analysis or any sort of partially observable model just an important question for discourse to continue on how is the Marco blanket restating the classical action perception cycle Ollie yeah just as a side note I wanted to point out again to um I think an important error in this particular figure which I've included in the Errata so if yeah and added a note there uh so if you could please open uh yeah for this yeah and open the notes there I've added the explanation for why exactly yes thank you so let's just put it what Giuseppe no it makes sense the the I think it's well spotted yes there's many changes and and um so again these are like why it's so important for everybody to make asynchronous contributions and questions if they can this is how we actually are going to move the needle on the textbook making sense and making impact I know it's like an individual sense making Journey as well but also these kinds of questions like what does this mean and the discourse around them is gonna catalyze the actual use of this area like um do you want to summarize

what the difference is between the existing equations and the replacement sure uh you see in the existing equation in the book The marginal flow of sensory states are expressed uh in terms of blanket States namely U and Y and also the um the inter the um I mean the external state but uh the it should be the other way around I mean the marginal flow sensory State should be expressed only in terms of external States and blanket States and the marginal flow of active States should be expressed in terms of are only internal States and the blanket States thank you so yeah just to restate that and also to build some intuition with what these flows are and how they relate so you and Y constitute B the blanket States so sometimes you and Y are both written or B is written to describe the blanket State variables other times it might just be describing a subset at which point not the whole blank is being described just for example the active States U or the sensory States y are being described then mu our internal States here and X you know find x hidden States external states with a note here that the variables are different in different presentations um as it was written we can read that this is the change in U which is the active States is a function of first a stochastic parameter about you itself but then the deterministic component here is a function of the blanket States and the hidden States and conversely as written why the change in observations Through Time is after accounting for a stochastic term a flow function on blanket States and internal States whereas it's actually exactly the opposite with X and μ being switched between these two because actions should reflect a flow over blanket and internal sometimes called the particular States because that's the particle that's moving around that's the Bayesian mechanical particle and then observations should reflect a flow on blanket States themselves and hidden states of the world and that's very related to this other question so uh what does figure 3.1 mean I think that's what we're looking at how is it different than figure two five and how is the agent's Niche relationship describe qualitatively and formally in active inference in other areas so I think we kind of explored um people can add any more notes but this is like a really important level to pull back to whether it's somebody who's having a or even uh well-intended or not criticism of active inference or a control theory it can be useful are they describing challenges of the world like the map territory difference like the complexity of the world always transcending the structure and complexity of our own models or referring to the fundamental issues of taking action amidst uncertainty are people pointing to those issues and then part of the community describing the agent niche qualitatively and or formally or are they sort of suggesting or implying some kind of like hidden other way of thinking about how agents engage in the niche because there's a lot of questions that can arise but these framings although here the Markov

blanket gives that kind of a Bayesian statistical and graphical uh stance on it and implementation opportunity space which includes an incredible number of computational affordances like message passing variational inference and so on partitioning of the entity from the niche figure from grounds is not diverging from the traditions of cybernetics and other areas so just kind of like interesting are people pointing to fundamental issues of perception and action which in some senses is our field and are they part of the building and the sense making around those scientific and meta-scientific models or is it just a position of like well it's not possible to model this kind of random thought but okay thank you yes Ali uh sorry just one other brief side note about the term you used there the particle particular States uh which you mentioned is the joint set of the blanket states and and the internal State the other related term used in the literature is the autonomous state which refers to the Joint set of internal and active States instead of the blanket States we only take the internal active States as the autonomous States or paths that's just this would kind of be a uh uh first and ask phrasing that the idea that these different um sets are going to inherit from the particular partition so the first move is the base graph to the particular partition which is just this specific partition between internal external States mediated by a Markov blanket then sometimes it's useful to talk about specific um sub sets of the particular partition if we were only interested in one of these categories we would just call it by name like talking about just external States or external State 11 dimension 11. um but other times we want to talk about partitioned States but still refer to more than one type of state and so Ali brought in this seconds clustering the particular States that's the particle that's moving around that's the whole entity internal States and blanket States and then also times it's important to consider the action in the internal States because those are kind of the ones that can be thought of as in a total control situation they're the ones that are under Direct Control so like you can control your mind and you can control your action those are the autonomous States whereas you can't directly influence sensory but you might be able to take actions that modify sensory states by changing where you look or by changing some aspect of the world um so those are autonomous States and it relates to this question can surprise only be diminished through action what does anyone think here's the throwback to yes Ali first yeah I just I wanted to point out to exactly that uh the same place in the text which I as far as I know there are two basic ways to uh reduce the surprisal or minimize the surprise one of which is uh through the action and the other one is through updating the uh generative model Yep this is the uh changed beliefs which is speaking more broadly than personally held convictions this is going to be pointing towards the Continuum

between perception learning and deeper belief States in generative models so perception is changing beliefs Bayesian beliefs which is to say updating the generative model learning and perception on the other hand there's action which can change the world and so out of this discrepancy it can be dissipated two ways it's kind of like if there was like a Delta G A Gibbs free energy like something that was um combustible there's like a hill to go down energetically and so now we're in an informational space but there's still at the um Clash of predictions and observations is a non-zero discrepancy and from that discrepancy there can be changes in the generative model parameters reflecting beliefs and also recourse to action and active inference is about how generative models which include beliefs about action resolve this kind of Nexus through the flows on updating oh sorry um yeah what is the difference between belief updating and mental action all added as a question and then someone can give a thought any thoughts well maybe mental action um refers to the active States of um and nested more more nested Mark of blanket so it's it's a it's an action it's a modulation of precision on the within the generative model and it's not a motor action you know through motor effectors states that actually communicate with the external world but it's still internal that's I don't know that's my take on it may be wrong yeah very uh nice I think just more more discussions can be added so just giving a quick thought belief updating could refer to any kind of parametric updating at all it doesn't even have to be like a cognitive model belief updating is just any time that there's a prior that meets with an observation and gets updated into a posterior you have belief updating and we can use Bayesian surprise to talk about belief then as pointed out mental action as it's used in the hierarchical models of it is referring to the policy selection process that's hinted in the action but the policy selection process or module or archetype as oriented around mental variables it is almost out of necessity requiring a nested or a hierarchical model because there has to be like a sort of base layer that is not mental action but then at the second level or above you can start to see the same kind of π policy variables and the same way of thinking about affordances and free energy and expected free energy minimization processes in the second level or above where those action like processes or maybe even genuine action processes are referring to cognitive States rather than a motor effector as would be understood as in the case of like conventional motor action whereas belief updating does not require a hierarchical model it's just referring very very broadly to Bayesian parametric change that makes sense thank you all right so can surprise be managed um through action no not only you're surprised you know you expect it to be 20 degrees um and you're observing it's 25. you can either engage in accent selection to change the thermometer under beliefs about how one's actions

change the world B Matrix or one can just update their model two that is now what I expect and of course that's the fun discussion in active inference with expect and prefer because if organisms are in the game of diminishing or bounding surprise then there has to be this nuanced dance between what is being observed understanding it to be simply the most likely but you don't want to be on the likely train off the cliff so there has to also be a space for slightly or very stubborn prior preference distributions that enable increasingly intense policies to be implemented to actually shape the state of the world and oneself back towards that earlier expectation and preference rather than engaging in just purely cognitive updating that's just speaking about it generally but there's a lot to um unpack in a more um model specific way okay I'll leave share the link oh awesome awesome this looks like a good reference and also uh live stream 25 I believe was on mental action Sanford Smith okay let's look towards this does anyone want to just give an example they can clarify synchrony between internal and external States what's the situation where we can think of internal and external State synchronizing well I guess it depends on what we mean by synchrony I mean synchrony like in uh in on a fast time scale or on a specific time scale because the fact that the internal State the reflect the the causal dynamics of the environment does not imply that they are immediately synchronized right yeah you're absolutely right the time scale really matters and also it's important to note that synchrony doesn't mean lockstep it doesn't mean that there's like a thermometer inside and a thermometer outside and they're just following each other um synchrony can refer to the kind of coordination that's more like orchestrated in the sense that if you had um two people you know sawing a tree they're both going to be taking opposite actions but there's still as high Mutual information between those two action policies and they're able to be under General synchrony um so we can just think of a few areas of examples like one can always think of a computer example because computers are clean like you could just say well I have two servers and they're communicating on the time or they're sending each other information back and forth or they're tracking some feature about each other you could imagine a communication example where there's some kind of semantic coordination or synchrony you could also imagine some kind of a cellular or a physiological synchrony like the inside of the cell is being modeled as doing inference about insulin levels outside of the cell and then the insulin receptor is like the interface and so there's synchronization of cellular response to perceived or inferred insulin concentration and the external unobserved true circulating insulin concentration mediated through set States The Binding kinetics of insulin on its receptors like these are just kind of qualitative examples but there's a lot to um drill

down into more Ali I think uh the word synchrony here uh might have been used in the technical term of uh the existence of a kind of synchrony map between the two coupling systems uh so in order for two systems to be coupled there needs to be a synchrony map uh namely Sigma between them and uh the whole paper worked example a paper by the alternate value well provides uh multiple uh physical examples for these kind of systems a couple systems and the most extreme case of um those couple systems are the sparse coupled systems in which the markup blanket uh is the strongest or as they call it the blanketedness or the blanket index uh is basically zero but in some other kinds of systems which in which the internal and external states are somehow more closely follow each other the markup blanket tends to weaken and diminish so yeah I think that's the meaning of the word synchrony here great Point yep live stream 49.1 with Jacob and Ali this week it's recommended listen and it's um quite far-ranging so that is the most state-of-the-art addressment of the issue a slightly older addressment of this issue and also um importantly coming back to what Ellie mentioned about like the synchronization manifold so here's the uh example of bird song that friston and frith have worked on and uh this is some of the colorized versions of the trace diagrams that are seen in the textbook that are like a little bit difficult to tell what's happening when they're on colored lines but just a highlight the manifold there the x-axis and the y-axis are the first and the second Bird's belief States and so to the extent that those pairs of beliefs at a given moment exist on the $Y = X$ line they're believing the same thing at the same time whereas if you could imagine if they were believing opposite things at the same time you'd have like a negative regression and if there was like uncorrelated or unsynchronized belief updating you'd have just a scatter Cloud that wouldn't have a significant regression slope and so this diagram is showing how before and after learning the sharpness of the manifold increases and a manifold is a lower dimensional space within a higher dimensional space so like we have a two-dimensional space here with X and Y and then there's activity only on this manifold which is being rejected effectively not completely down into one dimension because it's like a number line or you can imagine it was a three-dimensional space that was being projected into a two-dimensional manifold so manifold is just some lower dimensional structure in a higher dimensional phase space and so synchrony as Ali mentioned is referring um quite particularly technically to the um existence of synchronization manifolds and that is related to sparse coupling blanket index all these topics all right so in our last couple minutes we'll look at this question about 3.4.1 and then um over the coming week just chill enjoy the week also add a few other questions or discourse so that we can come back to some of the

latter sections of chapter three um next week so let's look at three four one in three four one that's the section variational free energy model evidence and surprise the authors look at the equivalence between variational free energy model evidence and surprise and formalize this relationship in equation 3.2 is it possible to unpack equation 3.2 to make these relationships easy to understand great question all right so let's go to equation 3.2 here's what the equation looks like then our um extremely helpful annotation group has gone through and provided this natural language unpacking so I'm just going to copy it down and then we can look at what it actually says and then see how that relates to the the question that was asked which was about unpacking the okay so the surprise fracture I fancy I of the data y conditioned vertical line on the model m is equal to the negative natural log $\ln$ of the conditional probability that's p of the data y given m with its upper bound being so this is a less than or equal to sign so they could be equal these two the first and the second one are equal the second one and the third one could be equal or the third term could be larger it's upper bound being the KL Divergence d_{kl} two lines the approximate posterior of hidden States Q of X and the conditional probability p of hidden States X conditioned vertical line on the data and the model minus the natural log of the model evidence P of the Y given the model evidence for the data given the model okay so the question was what are the relationships between that's the third term model evidence the second term and surprise the first term the answer is not in the simple reading of the equation but gets us to the discussion point so how are these terms related the three labeled terms on the bottom a related question is why do we use what is effective or useful about energy measures at all I'll give one thought and then one or two other people could add in which is we don't want to be outside of our range of surprise and so by defining surprise in terms of data um and specifically within the natural log space of data which has some just nice properties we want to stay bounded in the model evidence if we want to have high or the highest or at least adequate model evidence and that is equivalent to surprise we can see the same y given m now if we had that and had confidence without a doubt in our model and we knew how to calculate the likelihood function of the data then things would be easy and there are situations where things are easy like but there's other cases where we can bound our surprisal which is to say bound our model evidence evidence lower bound the elbow elbow and we can bound this intractable quantity model evidence with a divergence-based energy calculation and so we can do gradient descent variational inference message passing to block prints all these amazing things on the variational free energy and just know that at best we're going to converge to the model evidence so this is an optimization of an optimizable quantity of variational free energy

whereas model evidence is like if you know it you know it but if you don't you're kind of stuck like How likely is this book and it's super hard to understand how changing one word would change the overall likelihood distribution and so that motivates approximate heuristics for Bayesian inference especially those that are authorized by the sparsity of the data and the variational free energy measure allows us to bound surprise with tractable Optima optimizable function rather than needing to for example flesh out the total space of data likelihoods and model likelihoods which might be futile I'm sure there are other better ways to add things so one thing that like after reading it uh after reading about the after reading a book and after reading other article it I kind of had an Insight but I don't know if it's correct so that the old energy stuff of course comes from the physics and it's very legitimate to be a bit um a bit a bit confused because this formulation of uh with the Kullback-Leibler Divergence is a purely statistical quantity so it's understandable that somebody says what does energy have to do with that it seems to mean that the two con the the the high high road and the Lower Road converge here because when you minimize the on your hand you can see you can say that the organism maintains its forms and function by minimizing surprise and that is uh maximizing model evidence but on the other hand if you do this um you're gonna be able to compute the approximate posterior that a good approximate posterior about the states the q and x the Q of x in a sense here surprise minimization is also a way to compute a good posterior over the states all right great thanks for that no we'll return to this yes let's return to it pick up here and any other questions in next week so thank you see you all later